

**A reproducible GeoAI framework for explainable spatiotemporal analysis
of national missing-person registers: multi-algorithm clustering and
multilevel validation (Peru, 2019–2025)**

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15 **Abstract**

16 Missing-person report registers make it possible to describe territorial and temporal patterns,
17 but interpreting them requires distinguishing the recorded volume from its frequency relative
18 to the population. This study develops and applies a reproducible GeoAI framework to the
19 142,374 reports in Peru's National Registry of Information on Missing Persons (RENIPED,
20 2019–2025), organized as a panel of 26 territories by 84 months. The framework integrates
21 rates with official denominators, multi-algorithm clustering, spatial association and multilevel
22 validation. Rates ranged from 2.84 to 9.96 reports per 10,000 person-years; ten territories, led
23 by Tacna, Apurímac, Huánuco, Cusco and Junín, had rates above the rest of the country after
24 false discovery rate control, and the classification held in two sensitivity analyses. Metropolitan
25 Lima accounted for a third of the reports, yet its rate exceeded that of the rest of the country by
26 only 14%. Spatial autocorrelation was significant with the four nearest neighbors (Moran's $I =$
27 0.288 ; $p = 0.018$) but not with contiguity; after Benjamini–Hochberg adjustment, cold spots
28 were detected in the north, but no hot spots. Clustering did not show enough separation or
29 stability to support discrete categories, and four states descriptively summarize monthly
30 profiles. A specification multiverse showed that the clusters and the hot-spot corridor of an
31 initial analysis depended on specific analytical decisions. The framework provides
32 complementary information for assessing institutional needs, without estimating individual risk
33 or setting resource-allocation rules.

34 **Keywords:** Missing persons; GeoAI; Spatiotemporal clustering; Spatial autocorrelation;
35 Report rates; Multilevel validation; Administrative registers

36 **Highlights**

- 37 • Ten of 26 Peruvian territories have report rates above the rest of the country
- 38 • Metropolitan Lima holds a third of all reports but not the highest rate

- Evidence of spatial association depends on how neighbors are defined
- Four states summarize monthly profiles without demonstrating natural categories
- A specification multiverse shows how findings depend on analytical decisions

1. Introduction

The disappearance of people is a humanitarian and public-security problem with profound consequences for families and communities [1,2]. As one of the few systematic sources on the phenomenon, administrative registers of reports are a strategic resource for guiding search, prevention and the allocation of resources, provided that they can answer a concrete question: how does the recorded report rate vary across territories? In Peru, the National Registry of Information on Missing Persons (RENIPED), managed by the National Police [3], accumulated 142,374 reports between 2019 and 2025, and its release as open data now makes it possible to address this question at the national scale [4], in a context in which the Ombudsman's Office regularly warns about the number of reports of missing women [5].

Answering this question requires distinguishing the volume of reports from their frequency relative to the population. A densely populated territory can accumulate many records without having the highest rate, whereas a less populated one can record fewer reports and, at the same time, a higher rate. Metropolitan Lima illustrates this tension well: because it concentrates a third of the country's reports, it would top any ranking or map of counts. Geospatial artificial intelligence (GeoAI) offers suitable tools for this analysis, from unsupervised clustering of territorial series to spatial association indicators that flag hot and cold spots [6–9], although its application to missing-person registers is still incipient [10,11].

These tools, however, force analysts to make decisions that can alter the result, such as the unit of analysis, the features, the number of clusters, the spatial weights matrix or the way p-values are computed. The modifiable areal unit problem has already shown that aggregation

decisions alone can change spatial conclusions [12], and when each decision is made only once it becomes difficult to tell whether a pattern reflects the data or the analytical path chosen to examine them [13,14]. This risk is particularly high in registers organized as panels, which combine few territories with serially correlated months.

The central question of this study is how recorded missing-person report rates are distributed and evolve across Peru's territories, which monthly profiles can be described through clustering, and what evidence of spatial association they present. As a complementary dimension, we examine to what extent the interpretation of these patterns holds under different analytical decisions. The general objective is to develop and apply a reproducible GeoAI framework that integrates rate comparison, the characterization of spatiotemporal states and spatial association analysis, together with validation procedures that delimit the scope of its results. To this end, we pose five specific objectives:

- **(i)** to estimate territorial report rates and their ratios to the rest of the country, quantifying their uncertainty and examining their annual persistence;
- **(ii)** to characterize monthly profiles through features of relative behavior and prior level, and to compare the partitions obtained with different clustering algorithms;
- **(iii)** to assess the global and local spatial association of rates, considering alternative neighborhood definitions and adjustment for multiple comparisons;
- **(iv)** to examine the temporal transfer of the states by applying them to subsequent years and comparing them with the partitions obtained in those periods; and
- **(v)** to assess the stability and sensitivity of the findings under resampling, association with territorial identity and a defined set of alternative specifications.

The applied contribution is a joint characterization of the territorial differences, monthly profiles and spatial association of the reports recorded in Peru between 2019 and 2025, which distinguishes how much is recorded in each territory, how frequently relative to its population,

how a month behaves relative to its recent history, and whether values are associated with those of neighboring territories. The methodological contribution is an integrated and documented implementation of existing methods that pairs each component with checks of uncertainty, stability or sensitivity; its application to other registers requires further validation.

2. Related work

2.1. GeoAI and missing-person registers

GeoAI combines machine learning with spatial data to describe and predict geographic phenomena [6,15,16]. In public-safety research, a common workflow aggregates records into territory-period panels, clusters them with centroid-based algorithms and maps the resulting states. Missing-person data have entered this field only recently: Ruiz Reyes et al. reviewed the statistical and data-analysis techniques that support the search for missing persons [10]; Escobar et al. released a monthly municipality-level panel of disappearance cases and their outcomes in Mexico [11]; and Johnson et al. applied spatial clustering to missing-person records to locate community outreach events [17]. These studies open registers to large-scale analysis but also expose it to their properties: reports that are not persons, intermittent coverage and few territorial units.

2.2. Rates and the spatial structure of risk

Comparisons among territories of different size rely on per-capita rates. Because rates computed on small populations are more variable, the empirical Bayes adjustment of Assunção and Reis corrects for this difference when measuring autocorrelation [18]. Moran's I measures whether neighboring territories tend to have similar rates [19]; local indicators of spatial association (LISA) decompose it by territory [7], and the Getis–Ord G statistics identify concentrations of high or low values [8,9]. Inference for these local statistics relies on conditional randomization, which fixes the value of the territory under evaluation and randomly

draws its neighbors' values from the remaining values [7]. With few territories, this distribution is discrete: a territory with a single neighbor among 26 admits 25 possible configurations, so its two-sided p-value cannot fall below $2/25 = 0.08$. Moreover, because the test is repeated for every territory, a multiple-comparison correction is required, such as the false discovery rate (FDR) control of Benjamini and Hochberg [20].

2.3. Clustering and validation of the number of clusters

Internal indices such as the silhouette coefficient [21] summarize how well separated clusters are, but they cannot indicate that clusters do not exist, because they are undefined for a single cluster [22]. The gap statistic, by contrast, compares the dispersion of the data with that of structureless reference data and admits a single cluster as a valid answer [23]. Stability-based validation complements both criteria by assessing whether a partition reappears when the data are perturbed [24,25], although its outcome depends on the type of perturbation [26]. In panels, this dependence is decisive: resampling rows treats the months of a territory as independent, whereas block resampling preserves temporal dependence [27].

2.4. Association between clusters and territorial identity

In supervised learning, information leakage occurs when model development or evaluation uses information that would not be available under the intended conditions of application [28,29]. Clustering territorial registers raises a related but distinct issue: features that represent usual levels can produce clusters associated mainly with the territory of origin instead of distinguishing temporal situations shared across territories. Normalized mutual information can quantify this association [30], but it does not establish on its own whether the association reflects real territorial differences or properties of the representation used.

2.5. *Robustness to analytical decisions*

When many teams analyze the same data, their conclusions can differ considerably [14]. Multiverse analysis [31] and specification curve analysis [32] address this problem by running all reasonable combinations of decisions and reporting the full distribution of results. Although these methods have been applied mainly to regression models, this study extends them to clustering and local spatial statistics as part of the validation.

3. Data

3.1. *The missing-person register*

Reports are recorded in the National Police's Police Report System (Sistema de Denuncias Policiales, SIDPOL) and consolidated by the General Directorate of Information for Security of the Ministry of the Interior (MININTER), which publishes monthly aggregates on the National Open Data Platform under an ODC-By license [4]. We used the file covering January 2019 to December 2025, whose integrity we verified through its SHA-256 hash (beginning with c41ebcce). The file contains 77,331 rows, each corresponding to a cell defined by the district of occurrence, year, month, age group, sex and nationality, and records a number of reports that together total 142,374.

Three properties of the register condition its interpretation. First, the unit of record is the report rather than the person, so that someone reported more than once is counted more than once, a known feature of police data on missing persons [33,34], and the register does not state whether the person was found. Second, 1,197 rows (1,357 reports) have an unknown age and 83 rows (90 reports) an unknown sex; all were retained without imputation. Third, MININTER describes 2025 as a preliminary year [35], so 2024 is the last consolidated year. Annual totals ranged from 18,192 reports in 2023 to 26,331 in 2019 (2020: 18,481; 2021: 20,141; 2022: 18,862; 2024: 19,420; 2025: 20,947).

3.2. Territories and population

Because the register codes the department of occurrence and separates Metropolitan Lima from the rest of the Lima department, we worked with 26 territories: 23 departments, the Constitutional Province of Callao, Metropolitan Lima (Lima Metropolitana, province 150100) and Lima Region (Región Lima, department 150000 minus province 150100). Population data come from the projections of the National Institute of Statistics and Informatics (INEI) at 30 June of each year [36], which INEI itself labels as preliminary. Between 2019 and 2025, the 26 territories totaled 233,303,895 person-years.

3.3. Coverage gaps

Eight department-month combinations have no rows in the source. Six correspond to January 2023 (Madre de Dios, Moquegua, Piura, Puno, San Martín and Tacna), a gap consistent with a press release of the Ombudsman's Office stating that police statistics as of 13 February 2023 did not include those departments [37]; the remaining two correspond to Moquegua in May and June 2020, and their coverage could not be verified. The eight cases were treated as missing data and never as zeros, so that 2,176 of the 2,184 department-months are observed.

We therefore used two definitions of the rate. The main definition divides reports by the person-years of the observed months (observed exposure), whereas the sensitivity definition uses the person-years of all months (total exposure). The national rate was 6.114 reports per 10,000 person-years under the first definition and 6.103 under the second.

The rate therefore expresses the frequency of recorded reports relative to the population and observation time of each territory. Because the aggregated data identify neither unique persons nor the outcome of each report, the rate does not estimate the individual probability of disappearance or the proportion of persons who remain missing, and territorial differences may reflect both the occurrence of the phenomenon and its reporting and recording, components that these data cannot separate.

3.4. *Panel, temporal splits and geometry*

The register was structured as a panel of 26 territories by 84 months that fully preserves the 142,374 reports in the source. For clustering and its temporal validation we defined two splits: in the main split, models are fitted on 2019–2023 and evaluated on 2024; in the secondary split, they are fitted on the consolidated period 2019–2024 and their transfer to 2025 is evaluated. Because the prior level (Section 4.2) requires at least six observed months among the previous 12, the first months of each series do not yield complete features; consequently, the main split contained 1,396 territory-month rows for fitting and 312 rows from 2024 for evaluation.

For the spatial analysis we started from a simplified layer of 197 provinces, derived from INEI census cartography and distributed by a third party [38]. Provinces were aggregated into the 26 territories of the panel, and the resulting layer was projected to UTM zone 18 South (EPSG:32718), a metric system suitable for computing distances and neighborhood relations. Under queen contiguity, in which two territories are neighbors if they share a boundary segment or point, each territory had on average 4.15 neighbors (range 1–7). As a sensitivity analysis we used the four nearest neighbors based on great-circle distance computed with the haversine formula.

4. Analytical framework

The framework comprises five stages that address objectives (i)–(v): rates and priority territories, spatiotemporal states, spatial structure, temporal dynamics and multilevel validation (Fig. 1). The first four stages start from the same territory-month panel, and the results of stages (ii)–(iv) are tested in the fifth stage through a set of checks and specification curves, so that every finding is accompanied by an assessment of its robustness against reasonable alternatives. To limit analytical flexibility, all parameters are fixed in version 3.3.2 of an analysis plan whose

SHA-256 fingerprint is checked at the start of every run, and the plan's history documents each change and its rationale, including the addition of the first stage after intermediate results had been reviewed.

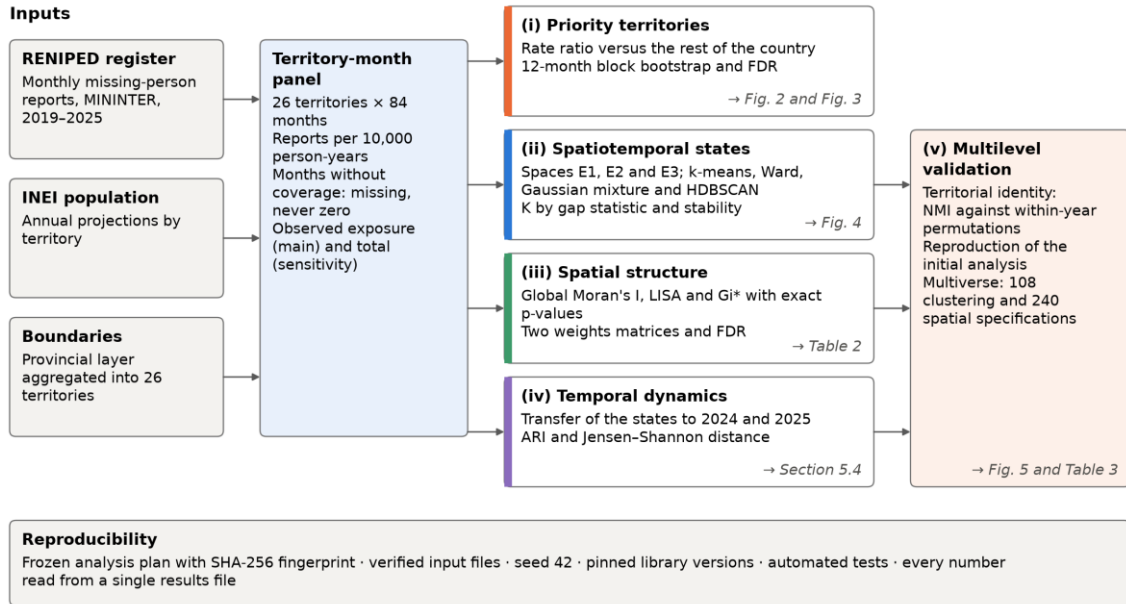


Fig. 1. Architecture of the GeoAI framework. The three sources are integrated into a panel of 26 territories by 84 months, from which four analytical stages start: (i) priority territories, (ii) spatiotemporal states, (iii) spatial structure and (iv) temporal dynamics. Stage (v) validates the results of stages (ii)–(iv) through territorial identity, the reproduction of the initial analysis and a multiverse of specifications. Italic arrows indicate where each result is reported; the bottom band summarizes the reproducibility safeguards.

4.1. Rates and priority territories

For each territory u , we computed the report rate per 10,000 person-years and compared it with that of the rest of the country, that is, with the pooled rate of the other 25 territories, through the rate ratio:

$$RR_u = \frac{rate_u}{rate_{rest(u)}} \quad (1)$$

A rate ratio of 1.5 means that the recorded rate of the territory is 50% higher than that of the other 25 territories combined over the period analyzed; it refers to reports per unit of population and time, not to a 50% higher individual probability of disappearance. We used the rest of the country as the reference so that the reports and population of the territory evaluated do not enter both terms of the comparison, which is especially relevant for Metropolitan Lima,

whose weight in the national total would push its comparison with the national mean toward equality.

The uncertainty of RR_u was estimated with a moving-block bootstrap [27,39] with 9,999 replicates. In each replicate, blocks of 12 consecutive months were drawn with replacement (seven for 2019–2025 and six for 2019–2024), the same for all 26 territories, and the reports and exposure of those months were used to recompute each territory's rate, the rate of the rest of the country and their ratio. Each block spans a full annual cycle, and joint resampling preserves the temporal correspondence between territories, although not necessarily the dependence between blocks. The 95% confidence intervals are the 2.5th and 97.5th percentiles of the replicates; they are individual intervals, not adjusted for the 26 comparisons.

To test $RR_u = 1$, we computed approximate two-sided p-values from the bootstrap distribution: we doubled the smaller of two proportions—replicates with a ratio ≤ 1 and replicates with a ratio ≥ 1 —each computed as $(\text{count} + 1)/(9,999 + 1)$, with a maximum of 1. We then applied the Benjamini–Hochberg procedure to the 26 comparisons [20]. A territory was classified as having a **higher** or **lower** rate than the rest when its adjusted p-value (q-value) was below 0.05, according to the direction of the estimated ratio, and as **not distinguishable** otherwise, that is, when no significant difference was detected. As a complementary persistence criterion, we counted in how many of the seven years its annual rate exceeded that of the rest, and we repeated the analysis, as a sensitivity check, for the consolidated period 2019–2024 and with total exposure.

4.2. Features and feature spaces

To describe the monthly evolution of each territory, we constructed the features in Table 1. Let $r_{u,t}$ be the monthly report rate of territory u in month t , that is, the reports recorded divided by the projected population for that year and multiplied by 10,000. To reduce the skewness of its distribution, we used the **transformed monthly rate** y :

$$y_{u,t} = \ln(1 + r_{u,t}) \quad (2)$$

The **prior level** N summarizes recent history as the mean of y over the previous 12 months, excluding the current month, using observed months and requiring at least six:

$$N_{u,t} = \frac{1}{|M_{u,t}|} \sum_{i \in M_{u,t}} y_{u,t-i} \quad (3)$$

where $M_{u,t} = \{i : 1 \leq i \leq 12 \text{ and } y_{u,t-i} \text{ observed}\}$ is the set of previous months with coverage, and $|M_{u,t}| \geq 6$ is required. The **change from prior level** C is the difference between y in the current month and that level: a positive value places the month above its recent history and a negative value below it, on the transformed scale rather than as a percentage change:

$$C_{u,t} = y_{u,t} - N_{u,t} \quad (4)$$

Finally, the **seasonal anomaly** A compares each month with the mean of the same calendar month in the same territory:

$$A_{u,t} = \frac{y_{u,t} - \mu_{u,m(t)}}{\sigma_u} \quad (5)$$

where $\mu_{u,m(t)}$ is the mean of y in the same calendar month and territory (for example, all Januaries) and σ_u is the standard deviation of the residuals from those seasonal means, both estimated only with the training years so as not to use information from the future. A thus expresses how far a month departs from its seasonal reference in units of the territory's residual variability; a value far from zero is not, on its own, a statistically significant anomaly.

With these features we defined three spaces that differ in how much territorial level they contain: the **relative space (E1)**, with A and C ; the **mixed space (E2)**, which is the main space and adds N ; and the **level space (E3)**, with y and N . The features of each space were standardized with the mean and standard deviation of the training set, and the same parameters were applied to the evaluation observations. This scaling differs from the territorial normalization of A , so standardized profiles are interpreted relative to the training distribution.

Table 1. Features and feature spaces. Internal code names are included only to facilitate verification of the reproducibility package.

Feature	Symbol	What it measures	Code name
Monthly rate	r	Reports per 10,000 inhabitants during the month	TASA_10K
Transformed monthly rate	y	Natural logarithm of $1 + r$	LOG_TASA
Prior level	N	Mean of y over the previous 12 months, with at least six observed months	NIVEL_PREV12
Change from prior level	C	Difference between y in the current month and N	CAMBIO_12M
Seasonal anomaly	A	Difference from the seasonal mean of y , divided by the territory's residual standard deviation	ANOMALIA
Territorial share of the register	P	Share of reports (panel) or rows (cells) of the territory; used only in sensitivity specifications	DPTO_ID
Space	Features included		
Relative (E1)	A, C		E1_relativo
Mixed (E2, main)	A, C, N		E2_relativo_mas_nivel
Level (E3)	y, N		E3_nivel

4.3. Multi-algorithm clustering

We clustered the months of each territory with k-means for $K = 2$ to 8 clusters, implemented in scikit-learn [40] (seed 42; 10 initializations during the selection of K and 20 in the final partitions). To check that the states did not depend on a particular algorithm, we also fitted Ward's hierarchical clustering, a Gaussian mixture model and HDBSCAN [41], and compared their partitions with that of k-means using the adjusted Rand index (ARI), which equals 1 when two partitions coincide and is close to 0 when they agree only by chance [42]. Clusters were numbered by their mean monthly rate, from lowest to highest, so that state 0 is always the lowest.

The evaluation distinguished between evidence of cluster separation and the selection of a partition to describe the data. For the former, we computed the gap statistic from $K = 1$, with 100 reference samples and Tibshirani's one-standard-error rule [23]. For the latter, we evaluated partitions with $K = 2$ to 8 by their stability, measured as the mean ARI over 100 resamples of whole territories and of 12-month blocks; row resampling is reported for comparison but does not enter the selection. Candidates were the values of K with a mean ARI of at least 0.80 under

both schemes, a threshold adopted as an operational criterion; among them, the rule chose the K given by the gap rule applied from $K = 2$ or, if that value was not a candidate, the one with the highest silhouette. If no K reached the threshold, a fallback rule chose the partition with the highest minimum stability across both schemes (the smallest K in case of ties), a descriptive summary that does not constitute evidence of separated clusters. We also report the silhouette and the Davies–Bouldin [43] and Calinski–Harabasz [44] indices.

4.4. *Spatial structure*

We measured the global autocorrelation of the 2019–2025 rates with Moran's I [19], with 9,999 permutations and two-sided p-values. To assess the robustness of the pattern we repeated the computation (a) with the consolidated period 2019–2024, (b) with the empirical Bayes adjustment, which stabilizes the variance of rates in sparsely populated territories [18], (c) with report counts and (d) with the proportion of months that each territory spent in the highest-rate state. A leave-one-out analysis also recomputed the index after removing each territory. At the local level we computed the Getis–Ord G_i^* statistic [8,9] and the local Moran's I [7] with PySAL [45].

For the main local analyses, with contiguity and with the four nearest neighbors, we obtained p-values by exact conditional randomization. With the value of the territory evaluated and its number of neighbors fixed, we enumerated all combinations of values that could occupy the neighboring positions among the other 25 territories; with the uniform weights used, both statistics depend only on the sum of those values. The two-sided p-value doubles the smaller of the proportions of combinations whose sum is greater than or equal to, or less than or equal to, the observed sum, with a maximum of 1, and in the absence of ties its smallest attainable value is:

$$p_{min} = \frac{2}{C(25,k)} = \frac{2 \cdot k! \cdot (25-k)!}{25!} \quad (6)$$

where $C(25, k)$ is the number of ways of choosing k neighbors out of 25. For a territory with a single neighbor, $p_{\min} = 2/25 = 0.08$, so Callao and Tumbes cannot reach significance at 5% with this test and the contiguity matrix; this restriction reflects the resolution of the procedure and does not imply an absence of spatial association. When the combinations exceeded one million, we used 99,999 random draws with a fixed seed. We controlled the FDR across the 26 territories with the Benjamini–Hochberg procedure, and we also report uncorrected results and, as a sensitivity check, results with the Benjamini–Yekutieli procedure, which is valid under any form of dependence among the tests [46]. Exact enumeration removes sampling variability, and the fixed seed makes the approximate computations reproducible.

4.5. Temporal dynamics

Temporal transfer was assessed by applying the model fitted in the training period to the observations of the following year. On the same observations, we refitted k-means with the same number of states and compared both assignments with the ARI. Because the refitted partition is an internal reference rather than a true classification, the ARI measures agreement, not predictive accuracy. We also compared the proportion of observations in each state between training and the evaluation year with the base-2 Jensen–Shannon distance [47,48], which ranges from 0 to 1 and describes changes in the distribution of states but not in individual assignments.

4.6. Multilevel validation

The last stage assesses whether the previous patterns reflect the phenomenon or depend on analytical decisions, and it does so at three complementary levels.

Association with territorial identity. This evaluation examines the extent to which the states obtained are linked to the territory from which the observations come. To this end, we computed the normalized mutual information (NMI) between states and territories [30] and the fraction of the uncertainty about the state that is removed by knowing the territory:

$$U = 1 - \frac{H(G|T)}{H(G)} \quad (7)$$

where $H(G | T)$ is the entropy of the states conditional on the territory and $H(G)$ is the entropy of the partition. A value of $U = 0.25$ means that knowing the territory removes a quarter of the uncertainty about the state, without implying that the territory causes membership. The observed NMI was compared with a reference distribution of 999 permutations that, within each year, randomly reassign territorial identities using the same reassignment for all months of each territory, which preserves the sequences within each year but breaks the territorial correspondence across years. We call **identity leakage** the operational signal defined by $p < 0.05$ and $U \geq 0.25$; this threshold organizes the comparison across specifications, but it is not a universal criterion and does not by itself demonstrate information leakage.

Reproduction of the initial analysis (baseline). To obtain a reference point, we reproduced an initial analysis of these data conducted by the authors in a conventional way. That analysis clustered the register cells into three states with nine features, robust scaling and the maximum-silhouette rule, and it reported a southern Andean corridor of hot spots using five nearest neighbors on hand-digitized centroids, one-sided p-values and no multiple-comparison correction.

Multiverse. To examine the sensitivity of the conclusions to analytical decisions, we evaluated a defined set of specifications [31,32], which does not exhaust all possible decisions. The clustering multiverse (M1) crossed the unit of analysis (panel or cell), the trimming of extreme values, the feature space (E1, E2, E3, E2 plus territory weight and, for cells, the nine initial features), the scaling and the rule for choosing K , for a total of 108 specifications. To keep execution feasible, M1 used a reduced budget: 20 resamples, five initializations, K between 2 and 6, 199 permutations for the null distribution, silhouettes on 5,000 rows and, as in the initial analysis, a subsample of 20,000 cells. The spatial multiverse (M2) crossed the trimming of extremes, the period, the variable (two rates or the count), five neighborhood

matrices (including the two of the initial analysis), the p-value type (one- or two-sided) and the multiple-comparison correction, for a total of 240 specifications. In each specification we evaluated claims defined in the plan, such as “there is structure according to the silhouette (≥ 0.50)”, “there is identity leakage” or “there is a southern Andean hot spot”, and we summarized the association between each decision and the metrics with partial eta squared (η^2_p), which we use only to rank the relative importance of factors within the design, not as a causal effect or an additive share of variance.

4.7. Reproducibility

The procedure was implemented in Python as a modular package, accompanied by a Google Colab notebook and ten automated tests of its critical components. The version analyzed (3.3.2) fixes all decisions in a versioned analysis plan whose history documents each change and its rationale. All random operations use seed 42, each input file is identified by its SHA-256 hash, and every analytical result is read from a single results file, from which the figures are also generated. The reference run on Google Colab (two processors; Python 3.13, NumPy 2.1.3, scikit-learn 1.6.1 and esda 2.9.0) took 30 minutes. A new full run on another computer, from the same input files and with the same versions, reproduced 34 of the 35 tables with relative differences below 10^{-10} ; the remaining table differed only in internal HDBSCAN indices that the article does not report.

5. Results

5.1. Rates and priority territories

The report rate varied 3.5-fold across territories, from 2.84 per 10,000 person-years in Cajamarca to 9.96 in Tacna (Fig. 2). Ten territories had a rate higher than that of the rest of the country, eight had a lower rate and eight were not distinguishable from the rest, a classification

that remained exactly the same under both sensitivity analyses, with the consolidated period 2019–2024 and with total exposure.

Five territories in the center and south of the country topped the list: Tacna ($RR = 1.64$; 95% CI: 1.27–1.96), Apurímac (1.58; 1.51–1.76), Huánuco (1.51; 1.42–1.65), Cusco (1.49; 1.40–1.58) and Junín (1.43; 1.31–1.52). In all of them, the annual rate exceeded that of the rest of the country in each of the seven years of the study (Fig. 3). They were followed by Ayacucho (1.32; 1.09–1.59; five of seven years), Lambayeque (1.26) and Arequipa (1.20), the latter two also above the rest in all seven years, and then by Metropolitan Lima (1.14) and Callao (1.06).

Metropolitan Lima clearly illustrates the difference between volume and intensity. Although it ranked first in number of reports (46,848, or 33% of the national total), it ranked ninth by rate (6.70), and its excess over the rest of the country, 14% ($RR = 1.14$; 95% CI: 1.05–1.19), recurred in six of the seven years. It is therefore a moderate and persistent excess, far from what its volume would suggest. Callao, with a rate of 6.48, showed an even smaller excess (6%).

At the opposite end, the eight territories with lower rates were Cajamarca ($RR = 0.45$), La Libertad (0.59), Piura (0.61), Loreto (0.61), San Martín (0.68), Áncash (0.73), Ica (0.77) and Amazonas (0.80); none exceeded the rest of the country in any year, and six of them (Amazonas, Cajamarca, La Libertad, Loreto, Piura and San Martín) are located in the north. The territories that were not distinguishable include some with wide intervals owing to their smaller populations, such as Madre de Dios (1.10; 0.75–1.26) and Pasco (0.99; 0.84–1.24), and others whose rates are close to that of the rest, such as Moquegua (0.94) and Puno (0.86).

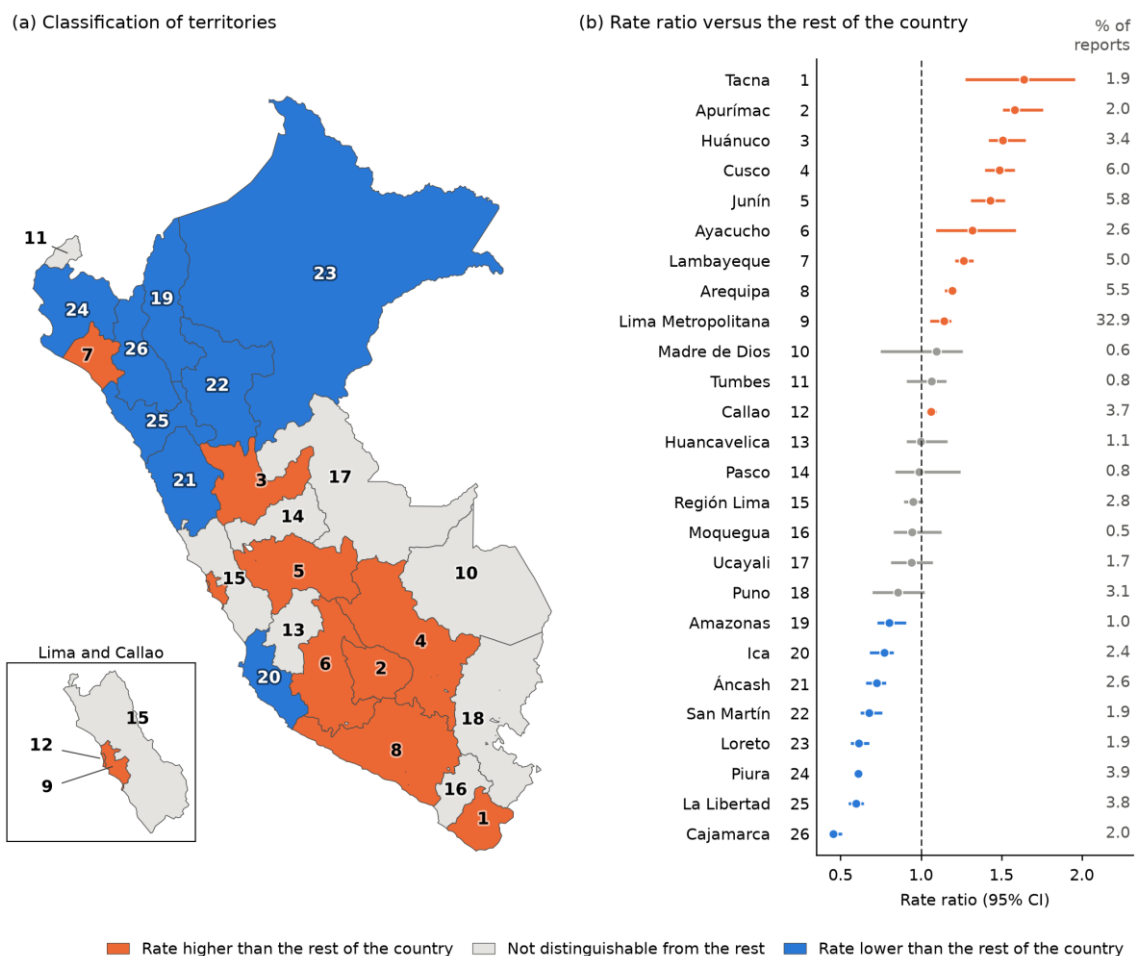


Fig. 2. Priority territories according to the missing-person report rate, 2019–2025. (a) Classification of the 26 territories: rate higher than, lower than or not distinguishable from that of the rest of the country (false discovery rate control at 5%); numbers identify each territory by its rank in (b), and the inset enlarges Metropolitan Lima (9), Callao (12) and Lima Region (15). (b) Rate ratio of each territory versus the rest of the country, with its 95% confidence interval (12-month block bootstrap, 9,999 replicates); on the right, each territory's share of national reports. Map lines delineate study areas and do not necessarily depict accepted national boundaries.

Fig. 3 breaks this classification down by year and makes it possible to distinguish a persistent excess from one concentrated in a few years. Apurímac, Huánuco, Cusco and Junín remained between 1.19 and 1.79 in every year, and the excess of Cusco grew from 1.29 in 2021 to 1.67 in 2025. In Tacna, by contrast, the excess declined steadily from 2.48 in 2019 to 1.01 in 2024 and rebounded slightly in 2025 (1.17), so its first place is driven mainly by the early years of the period. Ayacucho exceeded the rest until 2023 but fell below it in 2024 (0.94) and 2025 (0.73). Territories with lower rates, by contrast, remained so consistently; Cajamarca, for example, never exceeded 0.54, whereas among the non-distinguishable territories Madre de Dios fluctuated between 0.45 and 1.65, which explains the width of its interval. These annual

ratios are descriptive and carry no confidence interval; the classification is based on the full period.

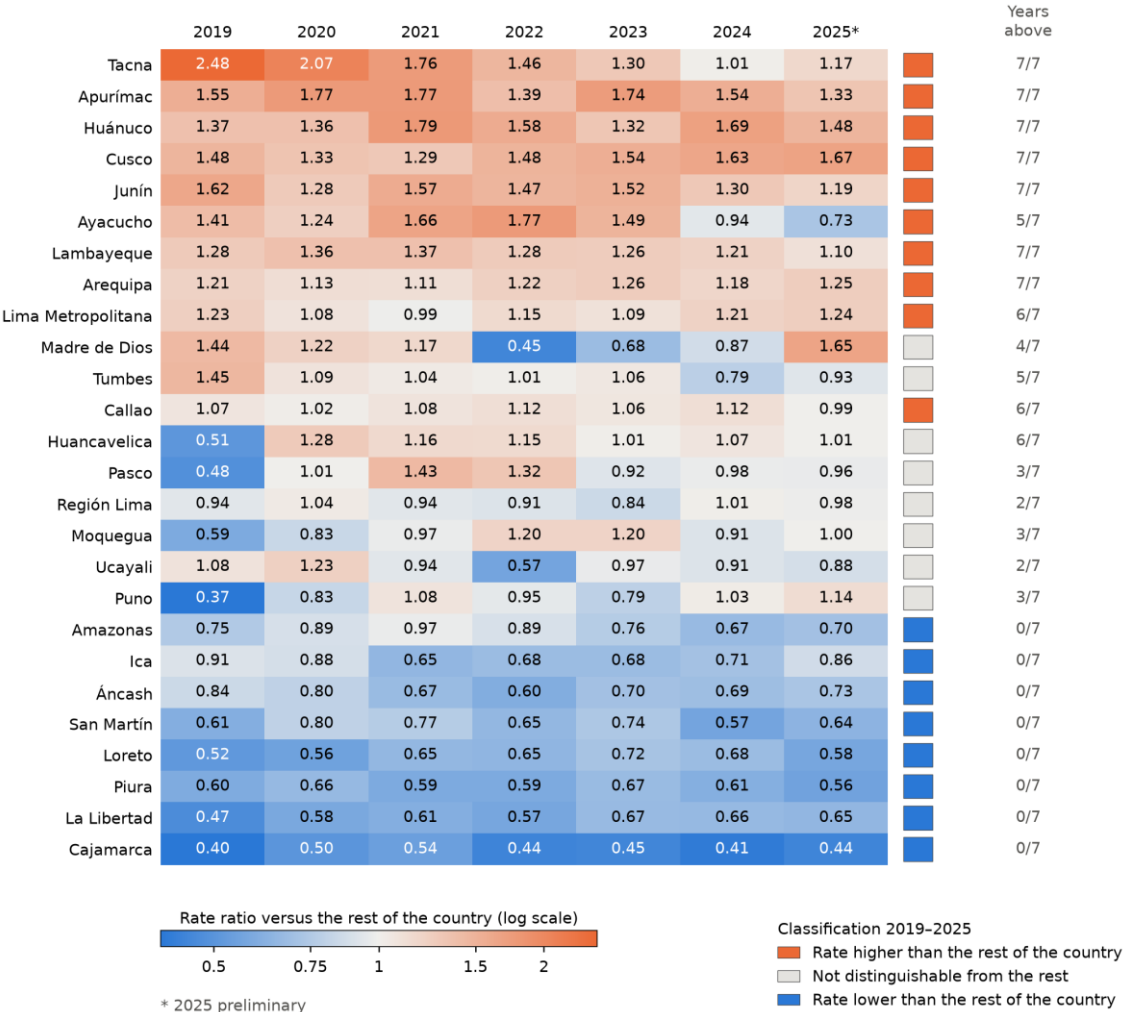


Fig. 3. Persistence of the rate ratio, 2019–2025. Each cell shows the ratio between the territory's rate in that year and that of the rest of the country (observed exposure); values above 1 (orange) indicate a higher rate, and values below 1 (blue) a lower rate. The color scale is logarithmic, so that 0.5 and 2 are equidistant from 1. The right-hand columns show the classification for the full period (Fig. 2) and the number of years in which the rate exceeded that of the rest of the country. Territories are ordered by their full-period rate ratio; 2025 is preliminary according to the source.

5.2. Spatiotemporal states

In the mixed space E2, the diagnostics provided no consistent evidence of clearly separated and stable clusters (Fig. 4a–c). The gap statistic favored $K = 1$, with a value of 1.851 compared with 1.380–1.432 for $K = 2–8$; the silhouette remained between 0.28 and 0.29, and no partition reached a mean ARI of 0.80 under both resampling schemes. The fallback rule was therefore applied, and it selected $K = 4$ as the partition with the highest minimum stability (0.63). The

440 four states are accordingly used to summarize monthly profiles of the features analyzed, not as
441 natural categories of the phenomenon. Agreement with other algorithms was also limited: the
442 ARI with respect to k-means was 0.31 for Ward and 0.23 for the Gaussian mixture, whereas
443 HDBSCAN labeled 70% of the observations as noise.

444 The profile of the four states reveals that they combine two types of difference (Fig. 4d).
445 Two of them describe the dynamics of the month and appear in all 26 territories: state 0 gathers
446 163 months well below the usual level, with a mean seasonal anomaly of -1.40 residual
447 standard deviations of the territory ($SD = 0.64$) and a mean monthly rate of 0.31 per 10,000
448 inhabitants ($SD = 0.16$), and state 3 gathers 294 months above the usual level, with an anomaly
449 of $+1.21$ ($SD = 0.55$) and a rate of 0.78 ($SD = 0.29$). The other two group months with anomalies
450 close to zero (-0.38 and -0.11) and differ mainly in the level of the territory: state 1 (500 months
451 from 19 territories) corresponds to territories with a low prior level, with a standardized median
452 of -1.00 , and state 2 (439 months from 20 territories) to territories with a high level, with a
453 median of 0.59 . This composition explains why, as shown in Section 5.5, the states share
454 information with territorial identity.

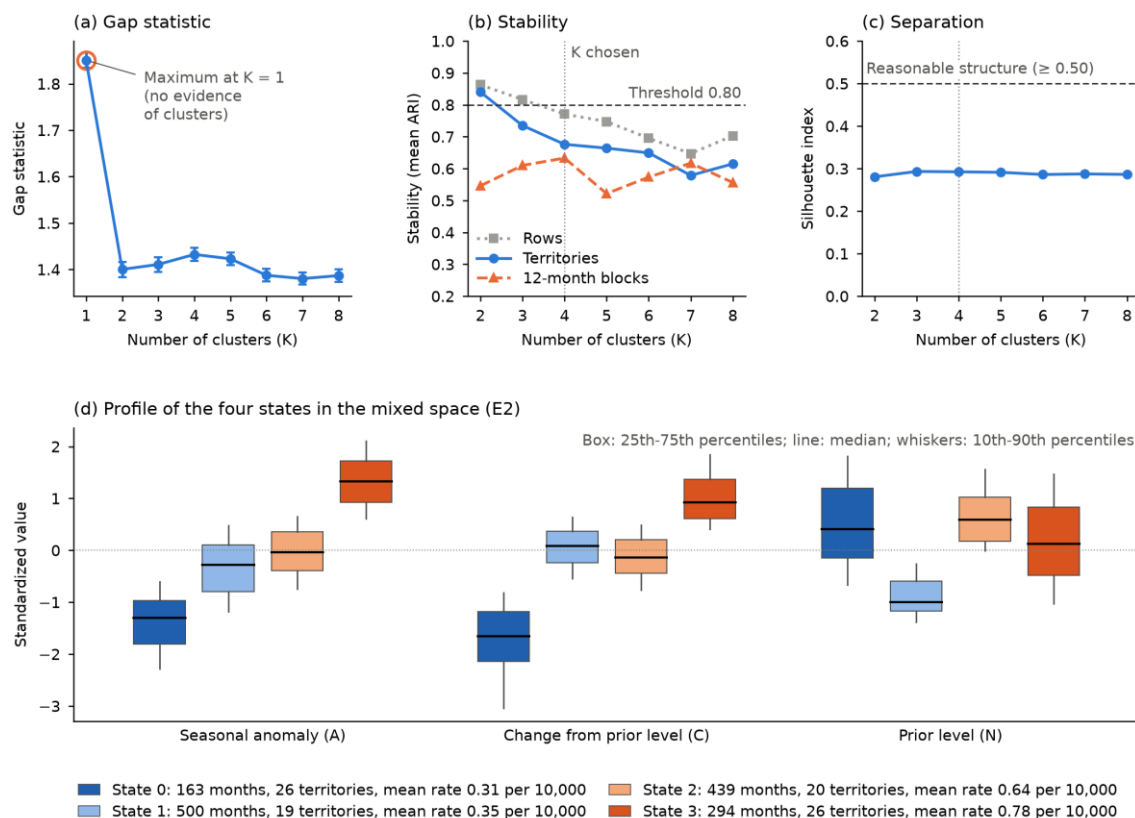


Fig. 4. Choice of the number of clusters and profile of the states in the mixed space (E2), 2019–2023 ($n = 1,396$ territory-months). (a) Gap statistic with its standard error; the diagnostic favors $K = 1$ and does not support a separation into several clusters in the representation evaluated. (b) Mean stability (ARI) over 100 resamples by resampling scheme; the dashed line marks the 0.80 threshold and the vertical dotted line the K chosen by the fallback rule. (c) Silhouette index; the dashed line marks the silhouette reference value adopted in the analysis (0.50). (d) Distribution of the three standardized features in each of the four states ($K = 4$); the legend gives the number of months, the territories present and the mean monthly rate of each state.

5.3. Spatial structure

Evidence of global spatial autocorrelation in the rates depended on the neighborhood definition (Table 2). With the four nearest neighbors, Moran's I was 0.288 ($p = 0.018$) for 2019–2025 and 0.245 ($p = 0.032$) for the consolidated period 2019–2024, whereas under contiguity it was 0.122 and not significant ($p = 0.244$); the empirical Bayes adjustment did not change these values. These results support a positive association under nearest-neighbor proximity, but not a conclusion independent of the spatial matrix. Counts and the proportion of months in the highest-rate state showed no significant autocorrelation with either matrix, and when each territory was removed in turn (24 of the 26 exclusions are computable), Moran's I under

contiguity ranged from 0.029 to 0.289 and was significant only when Huánuco was excluded, which shows its sensitivity to the territorial composition.

In the local G_i^* analysis, cold spots were identified after Benjamini–Hochberg adjustment in Amazonas and Lambayeque under contiguity, and in Lambayeque and San Martín with four neighbors; under the more conservative Benjamini–Yekutieli adjustment, only Amazonas remained significant ($q = 0.032$). No territory was a hot spot after adjustment with either main matrix. Under contiguity, Apurímac, Cusco and Ayacucho had the highest positive G_i^* values (z between 2.13 and 2.43), but their two-sided p -values (0.057–0.064) were not significant even before adjustment. Metropolitan Lima showed no significant local concentration either (under contiguity, $z = 0.07$ and $p = 0.97$).

The local classification differs from the comparison of each territory with the rest of the country. Lambayeque has a rate above that of the rest (Section 5.1) and, at the same time, a low-value surrounding that G_i^* identifies as a cold spot, whereas the local Moran's I characterizes it as a high-value territory surrounded by low values; both readings describe different aspects of its spatial position.

Table 2. Spatial structure of missing-person reports. Panel A: global Moran's I (9,999 permutations; two-sided p -value). Panel B: territories significant after false discovery rate control (exact local p -values; 2019–2025 rate with observed exposure).

Panel A. Variable and period	I (contiguity)	p	I (4 neighbors)	p
Rate, observed exposure, 2019–2025	0.122	0.244	0.288	0.018
Rate, total exposure, 2019–2025	0.116	0.260	0.282	0.020
Empirical Bayes rate, 2019–2025	0.122	0.244	0.288	0.018
Rate, observed exposure, 2019–2024	0.094	0.325	0.245	0.032
Report count, 2019–2025	–0.008	0.537	–0.004	0.451
Proportion of months in the high state, 2019–2023	–0.104	0.663	0.014	0.589
Panel B. Territory (matrix)	G_i^* z	Exact p	q (FDR)	LISA type
Amazonas (contiguity)	–3.01	0.0003	0.008	Low-low
Lambayeque (contiguity)	–1.95	0.002	0.023	High-low
Lambayeque (4 neighbors)	–2.12	0.002	0.037	High-low
San Martín (4 neighbors)	–2.84	0.003	0.037	Low-low

I : global Moran's index (expected value without autocorrelation: -0.04). G_i^* z: standardized value of the Getis–Ord statistic; negative values indicate a concentration of low values (cold spot). q: p -value adjusted with the Benjamini–Hochberg procedure; under the Benjamini–Yekutieli adjustment, only Amazonas (contiguity) keeps $q < 0.05$ ($q = 0.032$). LISA type: territory (high or low) relative to its neighbors (high or low). No territory was a hot spot after correction.

5.4. *Temporal dynamics*

In 2024, the partition transferred from the 2019–2023 fit had an ARI of 0.41 with respect to the partition refitted in that year; the proportion of observations in state 1 rose from 35.8% during training to 56.4%, and the Jensen–Shannon distance between the state distributions was 0.18. In 2025, with the model fitted up to 2024, the ARI was 0.52 and the distance 0.05, so the proportions moved closer to those of training, although differences between the transferred and refitted assignments persisted. These results describe partial temporal agreement and do not allow the states to be interpreted as categories that are invariant across periods.

5.5. *Multilevel validation*

Association with territorial identity. The information shared between states and territories increased as the features incorporated territorial level (Table 3, panel A). In the relative space (E1), the NMI was 0.015 and did not exceed what chance would produce ($p = 0.81$); in the mixed space (E2) it rose to 0.138 ($p = 0.001$), with 24% of the clusters' entropy explained by territory, just below the 0.25 threshold; and in the level space (E3) it reached 0.252, with 45% of the entropy explained and three territories whose months almost always fell in the same cluster. Requiring 12 months of history instead of 6 preserved the same ordering among spaces (NMI of 0.026, 0.166 and 0.271), although with that requirement E2 exceeded the leakage threshold (fraction 0.29).

Reproduction of the initial analysis. The framework reproduced the initial analysis: with register cells, nine features, robust scaling and the maximum-silhouette rule, it selected three clusters with a silhouette of 0.559 (0.550 in the initial report), and with its weights matrix it obtained a Moran's I of 0.330 and six hot spots (Arequipa, Ayacucho, Cusco, Moquegua, Puno and Tacna). That partition, however, shared a great deal of information with territory (NMI of 0.330; in 12 of the 26 territories, at least 95% of the cells fell in a single cluster), and with standard scaling the same specification selected two clusters, with each territory falling entirely

in one of them. The three initial clusters thus described mainly differences in territorial size and identity.

Multiverse. Across the 108 clustering specifications, row resampling indicated stability in 97% of cases, compared with only 31% when territories and blocks were resampled; the silhouette exceeded 0.50 in 34%, and 62% showed identity leakage according to the operational criterion of Section 4.6 (Table 3, panel B; Fig. 5a). Within the set evaluated, the feature space was the decision most associated with differences in NMI ($\eta^2_p = 0.60$), and scaling and the unit of analysis were the most associated with differences in silhouette ($\eta^2_p = 0.37$ each).

In the 160 spatial rate specifications, Moran's I was significant in 72.5% (65% with two-sided p-values), never under contiguity and, with two-sided p-values, in 50–100% of the specifications with the other four matrices (Fig. 5b). The southern Andean hot spot appeared in all specifications with uncorrected one-sided p-values, but in none with two-sided p-values and FDR control. In the latter scenario, only the inverse-distance matrix of the initial analysis flagged hot spots (8 of 40 specifications: Apurímac, Madre de Dios and Puno), with local p-values obtained by random draws because that matrix assigns each territory between 6 and 18 neighbors. Metropolitan Lima was not a hot spot in any of the 240 spatial specifications.

Fig. 5 summarizes both validations through specification curves. Each point represents a specification, ordered by the value of the metric, and the lower marks identify the options used; their concentration in one sector of the curve signals an association between the decision and the result, without isolating its effect on its own. In panel (a), 39 of the 41 specifications without identity leakage use relative (E1) or mixed (E2) features, and none uses only level features (E3) or the nine features of the initial analysis, whose specifications are among those with the highest NMI (0.26–0.33). In panel (b), contiguity occupies the eight lowest positions without any significant specification, whereas the five-neighbor matrix of the initial analysis sits at the top.

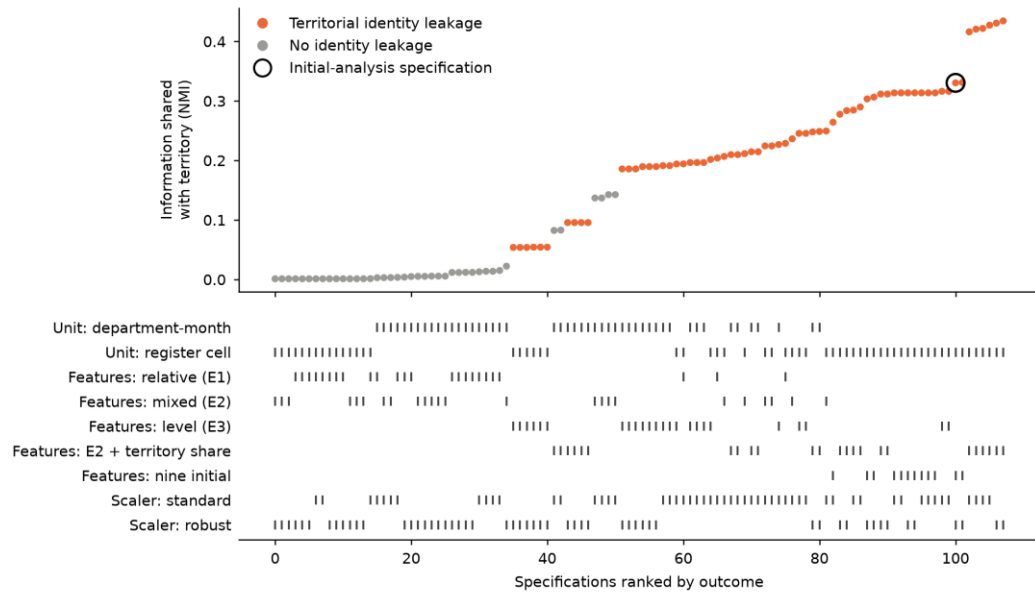
Table 3. Multilevel validation. Panel A: information shared between states and territories by feature space (2019–2023, $K = 4$; 999 within-year permutations). Panel B: proportion of multiverse specifications that support each claim.

Panel A. Feature space	NMI	Null threshold (p95)	p-value	Entropy fraction
Relative (E1)	0.015	0.021	0.813	0.027
Mixed (E2)	0.138	0.065	0.001	0.240
Level (E3)	0.252	0.116	0.001	0.453

Panel B. Claim	Specifications	Support
Structure according to the silhouette (≥ 0.50)	108	34.3%
Structure stable under row resampling	108	97.2%
Structure stable under territory and block resampling	108	30.6%
Territorial identity leakage	108	62.0%
Global autocorrelation of rates	160	72.5%
Southern Andean hot spot (one-sided p, uncorrected)	40	100%
Southern Andean hot spot (two-sided p and FDR)	40	0%
Hot spot in Metropolitan Lima	240	0%

NMI: normalized mutual information between cluster and territory. Entropy fraction: share of the uncertainty about the cluster that is removed by knowing the territory. Southern Andean hot spot: at least one of the five territories of the corridor described in the initial report (Arequipa, Ayacucho, Cusco, Moquegua and Tacna), as defined in the plan; Puno, the sixth hot spot in the reproduction, is reported separately. With two-sided p-values and FDR, 8 of the 40 specifications flagged other hot spots (Apurimac, Madre de Dios and Puno), all with the inverse-distance matrix of the initial analysis. Percentages are the proportion of specifications evaluated that meet each criterion; they are not probabilities that a claim is true and do not come from independent replications, because the specifications share data and analytical decisions.

(a) Clustering: information shared with territory (108 specifications)



(b) Spatial structure: global Moran's I of rates (40 specifications)

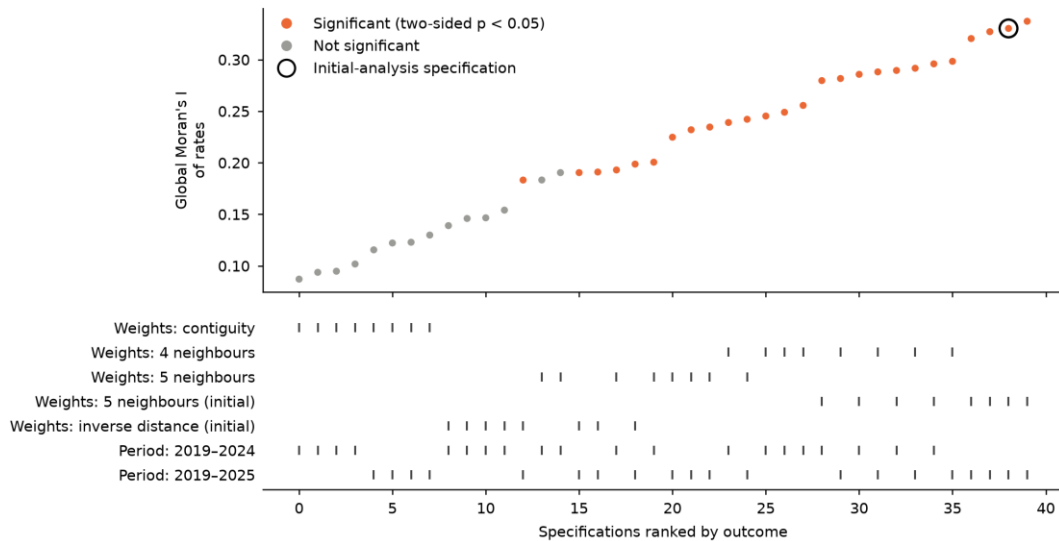


Fig. 5. Multilevel validation through specification curves. (a) Normalized mutual information (NMI) between cluster and territory in the 108 clustering specifications; orange marks specifications with territorial identity leakage ($p < 0.05$ and entropy fraction ≥ 0.25). (b) Global Moran's I of the rates in 40 combinations of data and weights matrix (two-sided p-values); orange marks those significant at 5%. In both panels, the circle marks the specification of the initial analysis and the lower rows indicate the decisions that define each specification. Only the most influential decisions are shown, because the omitted ones have little influence: in (a), the rule for choosing K ($\eta^2_p = 0.10$) and winsorization ($\eta^2_p < 0.01$); in (b), the exposure definition and winsorization, which change the mean Moran's I by less than 0.01.

6. Discussion

6.1. Territorial differences in report rates and implications for search

The result with the broadest policy relevance is the classification in Fig. 2. Ten territories have report rates above the rest of the country after FDR control, and five of them (Tacna,

Apurímac, Huánuco, Cusco and Junín) exceed it by 43% to 64% over the full period, with an annual rate above that of the rest in each of the seven years. The agreement of this classification across the two sensitivity analyses supports its consistency under the variations in period and exposure examined. These results can provide a complementary criterion for assessing needs for search, prevention and review of the register, provided they are interpreted together with the volume of reports, the status of cases and institutional response capacity; on their own, they do not establish a resource-allocation rule. Fig. 3 adds a useful nuance for planning: the excess of Apurímac, Huánuco, Cusco and Junín is stable, that of Cusco is increasing, and that of Tacna declined until it matched the rest of the country in 2024 (1.01) and rebounded only slightly in 2025. The classification should therefore be updated every year rather than setting priorities only from the period average.

Metropolitan Lima illustrates the value of considering the volume and the rate of reports together: the count describes the magnitude of recorded activity, whereas the rate places that volume in relation to the population and allows comparison with other territories. An institutional assessment of priorities should combine both dimensions with information on pending cases, their urgency and the available resources, aspects that lie beyond the scope of this study.

The low rates in the north, in turn, should be interpreted with caution, because a low report rate may reflect fewer disappearances but also poorer access to police stations, a lower willingness to report or differences in recording. The available data cannot discriminate among these explanations, so we do not attribute causes to the observed patterns. The same caution applies to high rates, which indicate where more reports are filed per inhabitant, but not why.

6.2. Spatial pattern and states of the phenomenon

The observed distribution shows high-rate territories in the center and south of the country, but interpreting it as spatial concentration requires distinguishing geographic description from

inferential evidence. Global autocorrelation was significant with the four nearest neighbors but not under contiguity, and no hot spot survived multiple-comparison adjustment with the main matrices, so the results do not confirm a hot-spot corridor under these specifications. Nor do they allow this absence to be attributed solely to insufficient power, since we did not assess power, although the smallest attainable p-value limits the resolution of the test in territories with few neighbors. In the north, cold spots were detected with Benjamini–Hochberg adjustment, but their location depended on the matrix and only Amazonas remained significant under Benjamini–Yekutieli, so local evidence should be read in light of the neighborhood and adjustment used.

The four states serve a descriptive purpose: they summarize combinations of seasonal anomaly, change from prior level and usual level, and they distinguish relatively low or high months from situations linked to persistent differences between territories. The limited separation and the sensitivity of the partitions, however, suggest interpreting their boundaries as analytical approximations rather than divisions inherent to the phenomenon. Rate classification and clustering also answer different questions: the former compares the recorded frequency over a period, and the latter summarizes multivariate monthly profiles, so the consistency of the former in two sensitivity analyses does not, on its own, demonstrate a general superiority over the latter.

6.3. What multilevel validation contributes

Multilevel validation helps to interpret the differences between the initial analysis and the alternative specifications. The initial clusters showed greater separation and stability under particular choices of unit of analysis, features and resampling, and their association with territorial identity indicated that part of that organization was linked to persistent differences between territories. This result does not by itself demonstrate information leakage, but it limits the interpretation of the clusters as shared temporal states. Likewise, the identification of the

southern Andean corridor changed when the sidedness of the test and the multiple-comparison adjustment were modified. Together, these results show the need to assess which conclusions hold under alternative analytical decisions.

Building on this experience, we recommend five checks for similar analyses: compare rates rather than counts, always with their uncertainty; evaluate the gap statistic starting from a single cluster; measure stability by resampling territories and time blocks; compute exact local p-values and report the minimum attainable p-value of each territory; and control the FDR with more than one weights matrix.

6.4. Limitations

The study is conditioned by the nature of the administrative register. The aggregated reports identify neither unique persons nor the outcome of cases, so rates describe the recorded frequency and not the individual probability of disappearance, and territorial differences may reflect both the phenomenon and access to reporting and recording procedures. Population projections and the 2025 records are preliminary, and the treatment of the eight territory-months without information adjusts the exposure but does not recover any reports that may be missing. Aggregation into 26 territories also prevents transferring the results to specific localities or persons.

Methodologically, the spatial findings depend on the neighborhood definition and on the multiple-comparison adjustment, and the geographic layer comes from a third party, so its correspondence with official cartography should be checked before assuming that contiguity does not change. The intervals of the rate ratios rely on fixed-length blocks over seven years, may be affected by changes in level or trend, and do not incorporate the uncertainty of under-reporting or of the projections. The states depend on the representation of the features, and their association with territorial identity, assessed with a specific annual permutation scheme, does not by itself separate persistent differences in the register from effects of that representation;

the temporal evaluation, limited to two years and without a true reference classification, does not measure predictive accuracy.

Finally, the framework was evaluated on a single register and without synthetic data of known structure, and the multiverse covers a finite set of decisions with a reduced computational budget. The territorial rate comparison was added to the plan after intermediate results were reviewed, as documented in its history, which gives that component an exploratory character. Applying the procedure in other contexts will require checking coverage, units of analysis and the assumptions of each method anew.

6.5. Future work

The study can be extended in three directions: evaluating a finer territorial scale, after checking coverage, denominators and the stability of rates in small areas; applying the procedure to other administrative registers to examine which components hold and which require adaptation; and running experiments with synthetic data representing temporal dependence, persistent territorial differences, coverage gaps and known spatial concentrations, in order to assess the calibration of the tests and the framework's ability to distinguish known patterns from variations introduced by analytical decisions.

7. Conclusions

This study integrated population rates, clustering and spatial analysis to characterize the 142,374 missing-person reports recorded in Peru between 2019 and 2025. Rates ranged from 2.84 to 9.96 reports per 10,000 person-years, and ten territories, led by Tacna, Apurímac, Huánuco, Cusco and Junín, had rates above the rest of the country, with a classification that held in both sensitivity analyses. Metropolitan Lima accounted for a third of the reports without having the highest rate, which shows the value of considering the recorded volume and its relation to the population together.

Spatial evidence depended on the neighborhood definition: global autocorrelation was significant with the four nearest neighbors but not under contiguity, and with the main matrices cold spots were identified after Benjamini–Hochberg adjustment, but no hot spots. In the clustering component, the diagnostics did not support clearly separated and stable clusters in the representation evaluated; the four selected states summarize monthly profiles, although their interpretation must take into account territorial association and the partial agreement between periods.

The contribution of the framework lies in articulating these perspectives and pairing them with checks that delimit the interpretation of the results. The findings can provide complementary information for assessing institutional needs, together with the status of cases and the available response capacity, but on their own they establish neither a resource-allocation rule nor an estimate of individual risk. The code, the analysis plan and the results are published to facilitate their verification; their extension to other registers requires validating the conditions of application and the behavior of the procedures used.

Data availability

The report data come from Peru's National Open Data Platform [4], the population denominators from the INEI projections [36] and the geographic layer from the peru-geojson repository [38]; the three files are identified by their SHA-256 hashes. These files, the code, the analysis plan, the automated tests, the Google Colab notebook and the derived results of version 3.3.2 are available on GitHub and archived in Zenodo [49].

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